Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/10/2006
PI Name:	LaCombe, Jeffrey C. Ph.D.		
Project Title:	The Study of Dynamics and Tip Selection in Thermal Dendrites via Pressure Moderated Step Changes in Supercooling		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Glasses and ceramics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Reno	**State:**	NV
Zip Code:	89557-0136	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	11/01/2003	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Paley, Mark	**Contact Phone:**	256 544-7775
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per D. Gillies (10/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Koss, Matthew (College of the Holy Cross) Glicksman, Martin (Rensselaer Polytechnic Institute) Chait, Arnon (NASA Glenn Research Center) Pines, Vladimir (NASA Glenn Research Center)		
Grant/Contract No.:	NNM04AA22G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The objective of this project, titled Tip Selection Dynamics via Pressure Change, is to study fundamental aspects of time-dependent dendritic growth, and obtain benchmark ground-based data on time-dependent and transient dendritic growth phenomena in single, isolated, dendrites. In order to make such measurements, rapid stepwise changes in pressure will be created, which will cause rapid changes in the growth conditions that are not otherwise possible. During these changes, we will closely observe the process in situ, compiling data on the manner in which the process evolves from one steady state to another. This proposed project grows from an earlier NASA-funded project, the Transient Dendritic Growth Experiment, which successfully passed its Science Concept Review, in preparation for flight aboard the International Space Station. Due to circumstances beyond our control (related to the availability of microgravity resources) we now seek to re-direct our activities towards further developing our understanding of the ground-based process, prior to pursuing a microgravity experiment. The investigator team includes the key personnel of the successful IDGE microgravity experiment, which brings over 40 years of cumulative experience in microgravity experiments. Our experience tells us that maximum return on microgravity resources occurs when the ground-based behavior of the process is understood to the fullest extent possible and the experimental apparatus is developed and tested completely. This proposed project will permit us to move forward in this direction, and continue the long-standing involvement of NASA in key crystal growth scientific studies and technology development. The knowledge that this proposed study will produce will improve our understanding of many industrially-important solidification processes by extending the important earlier results of the NASA-sponsored Isothermal Dendritic Growth Experiment (IDGE), by better revealing how dendrites develop over time. This is important in a complex (and difficult to study) environment such as a casting, where boundary conditions change over time.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	During the past year of this grant's activities, we made progress in developing a relatively inexpensive and flexible apparatus for performing dendritic crystal growth experiments with pressure changes. We developed, procured, and tested a hydraulic pressurization system to induce the pressure changes necessary in our experiments. Chamber Testbed A was used with our pressurization system (both developed in previous year) to develop software-controlled pressurization. The project's graduate student, Syam Nibhanupudi, spent a considerable amount of effort to develop Labview code that interfaces with data acquisition and control hardware to perform this function. During testing, it was found that the growth chamber, when coupled to the pressurization hardware, changed the dynamic behavior, making it necessary to develop new control algorithms. At present, we are able to implement rapid pressure changes over the entire desired range of pressures. We are having difficulties with maintaining constant pressures over several 10s of minutes, though, particularly at high pressures. We believe this is due to small amounts of hydraulic seepage in the system that have a measurable effect only over extended durations. We also experienced problems with pressure spikes upon initially changing the pressure (rapidly), although this problem was mitigated through the use of flow-control orifices in the hydraulic system. On the whole, we believe we are ready to conduct experiments over shorter durations (or periodic pulses in pressure), and we are confident that we will be able to address the long-term leakage through replacement of components, or through changes to the control software (more active feedback control) that correct for the gradual pressure losses. Chamber Testbed B is currently under development. This has been slowed due to the challenges experienced with controlling the pressure in Chamber Testbed A, when coupled to the pressurization system. The principal design changes will include the addition of optical windows capable of withstanding the imposed pressure differential. Testbed B will again be tested by filling with distilled water and tested under pressure. Chamber Testbed C development has been postponed until the performance of Testbed B is known. The principal design changes will permit testing and experimentation with SCN (rather than water tests). The Holy Cross team developed software tools for dendritic growth analysis, and performed significant educational outreach activities, offering an educator's workshop related to microgravity. 21 participants attended 12 full-day meetings, giving very positive feedback.
Bibliography Type:	Description: (Last Updated: 03/10/2006)
Articles in Peer-reviewed Journals	Koss, M.B. ; LaCombe, J.C.; Chait A.; Pines V., Zlatkowski, M.; Glicksman, M.E.; Kar, P. "Pressure-Mediated Effects on Thermal Dendrites" Journal of Crystal Growth , Jan-2005
Articles in Peer-reviewed Journals	Giummarra, C.; LaCombe, J.C.; Koss, M.B.; Frei, J.E.; Lupulescu, A.O.; Glicksman, M.E. "Sidebranch characteristics of pivalic acid dendrites grown under convection-free and diffuso-convective conditions" Journal of Crystal Growth , Jan-2005
Articles in Peer-reviewed Journals	Kar, P.; LaCombe, J.C.; Koss, M.B. "Velocity and Radius Transients During Pressure-Mediated Dendritic Growth of Succinonitrile" Materials Science & Technology , Dec-2004